Characterization, modelling and design of laser-textured surfaces with selective radiative properties

From processing to spectral selectivity: surface descriptors, simulation and physics-informed machine learning

ABSTRACT

The radiative properties of materials govern heat transfer by radiation, and their spectral selectivity is key for emerging technologies such as concentrated solar power (CSP) and radiative cooling. Direct laser interference patterning (DLIP) can produce spectrally selective surfaces, but their optical response is difficult to predict. It depends not only on the pattern geometry but also on the surface state left by the process, such as oxide layers. This project hypothesizes that the radiative behavior of DLIP-textured surfaces can be predicted from a reduced set of geometric and compositional descriptors. To test this, samples will be fabricated under varying processing conditions and characterized using the HAIRL emissometer, UV-VIS-NIR-MIR spectroscopy, and surface and compositional analysis. These measurements will link processing parameters to surface descriptors, and those descriptors to radiative properties. Rigorous coupled-wave analysis (RCWA) simulations that include the measured oxide layers will be combined with experimental data to train a physics-informed machine learning model. This addresses the gap between idealized simulations and real surfaces. Finally, the model will be validated by designing, fabricating, and measuring surfaces with a target spectral response. The expected outcome is a validated framework and a set of design rules for tailoring the spectral selectivity of laser-textured surfaces.

REQUIREMENTS

Nobody is expected to bring all of this. The degrees are the formal part; the rest is welcome and is taught along the way.

- Bachelor's degree in Physics, Materials Engineering or a related field, and a Master's degree in Materials Science, Physics, Nanotechnology or a related area, as required for admission to a doctoral programme (or expected completion before the start date).
- Welcome, not required: some background in optics and radiative heat transfer, or in materials characterization techniques (spectroscopy, microscopy or surface analysis); machine learning or numerical simulation (e.g., electromagnetic methods such as RCWA or FDTD); some programming, ideally in Python, for data analysis.
- Interest in experimental laboratory work and in linking measurement, simulation and design, willingness to work in a team, and English good enough to read, write and present scientific work (B2 or higher).

Full or partial funding is available for the thesis projects.

SUPERVISORS

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